



AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH

Faculty of Engineering

LabReport

Experiment # 07

Experiment Title: Interfacing the Arduino with an external sensor using serial communication protocol for implementing an obstacle detection system.

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Course Teacher:	DR. MD. RUKONUZZAMAN		

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Experiment Title: Interfacing the Arduino with an external sensor using serial communication protocol for implementing an obstacle detection system.

Objectives:

The objectives of this experiment are to-

1. Write code for a simple obstacle detection system in Arduino IDE.
2. Implement a simple obstacle detection system using an Arduino micro-controller.

Equipment List:

- 1) Arduino IDE (any version)
- 2) Arduino Uno (R3) board
- 3) Sonar Sensor (HCSR04)
- 4) Breadboard
- 5) LEDs (red, green, and yellow)
- 6) Resistors of 100Ω
- 7) Jumper wires

Circuit Diagram:

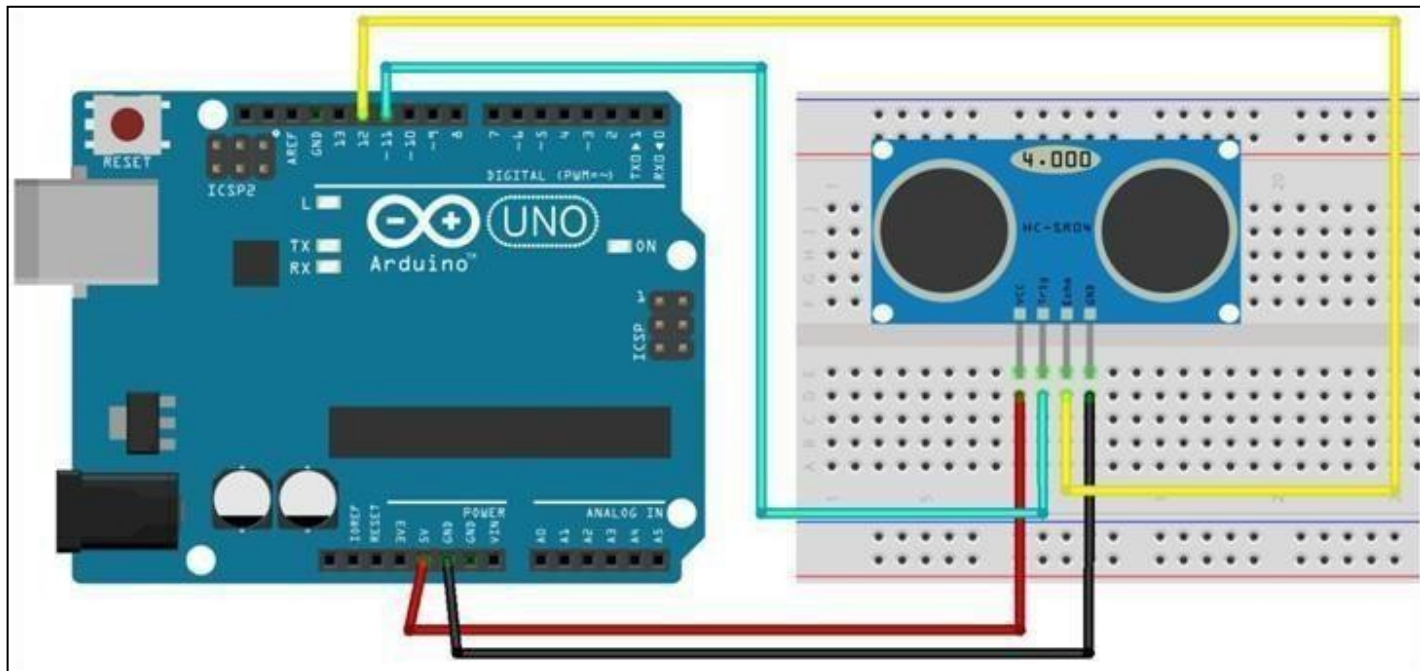


Figure-1: Arduino board's pin connections with the Sonar Sensors (schematic diagram)

Experimental Output Results:

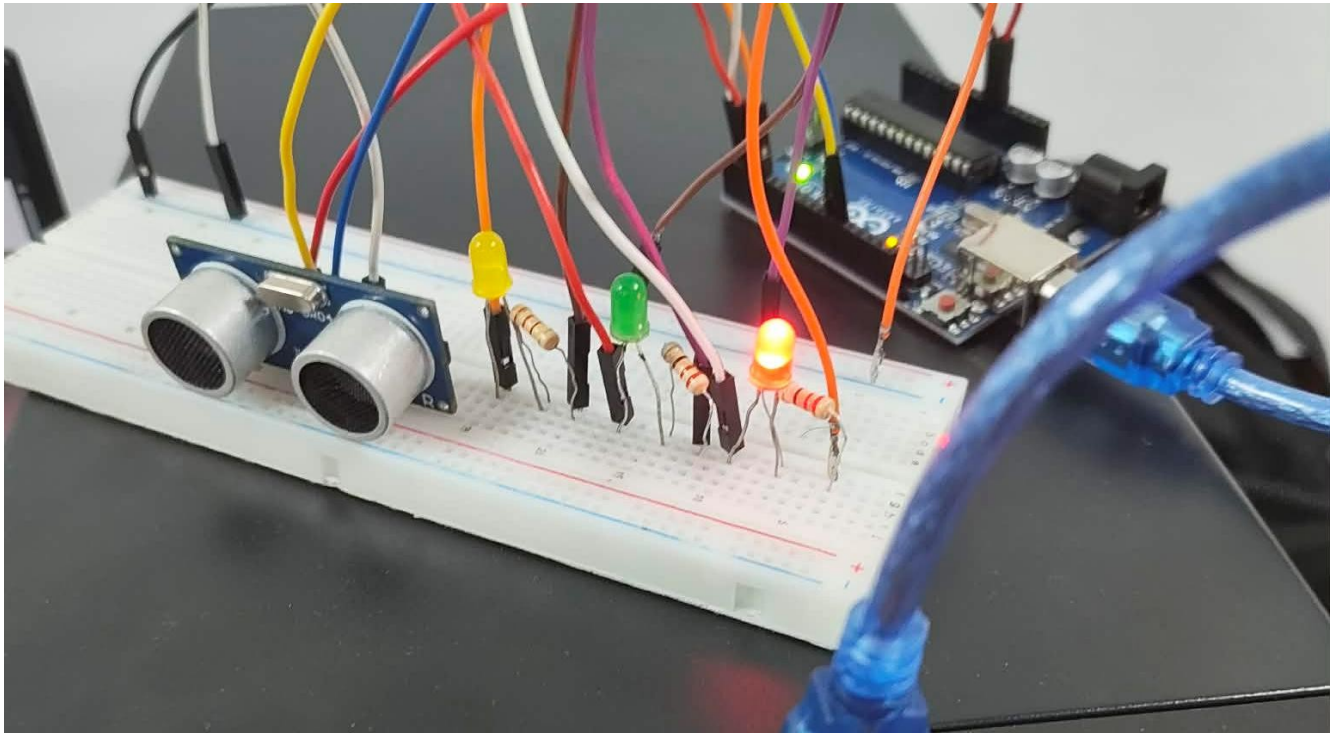


Figure 2: Distance > 80 cm **RED LED ON** only

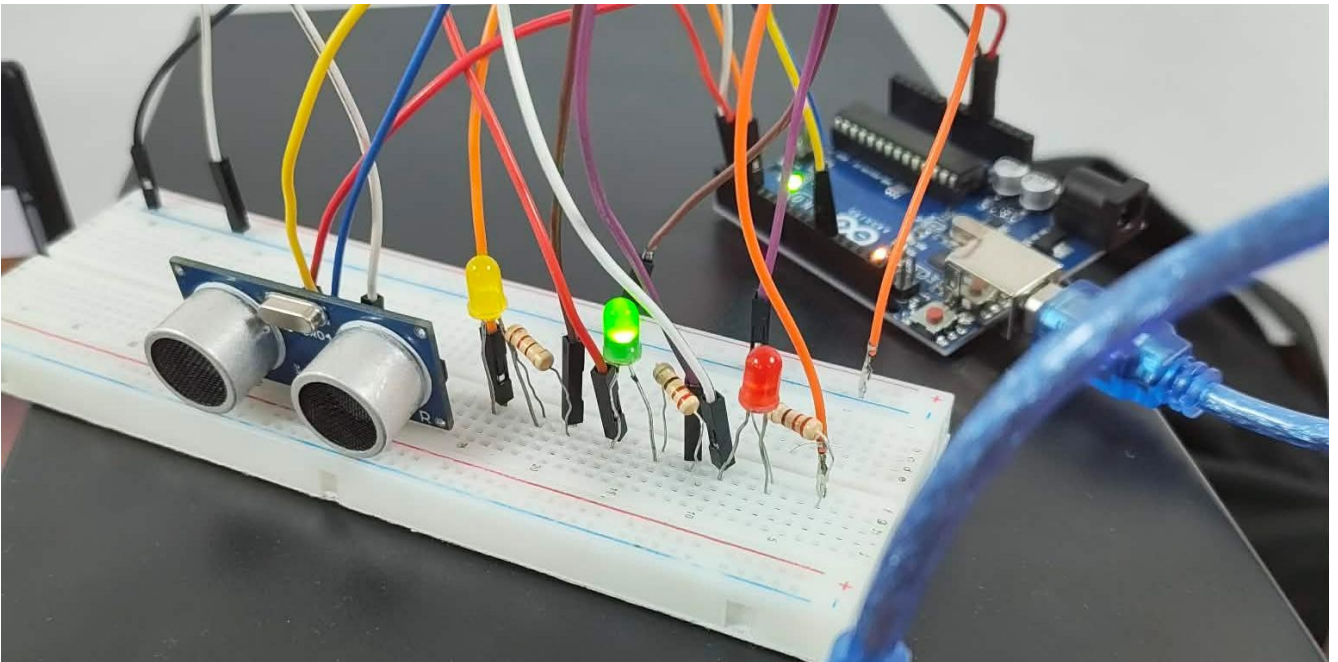


Figure 2.1: $50 \text{ cm} < \text{Distance} \leq 80 \text{ cm}$ **GREEN LED ON** only

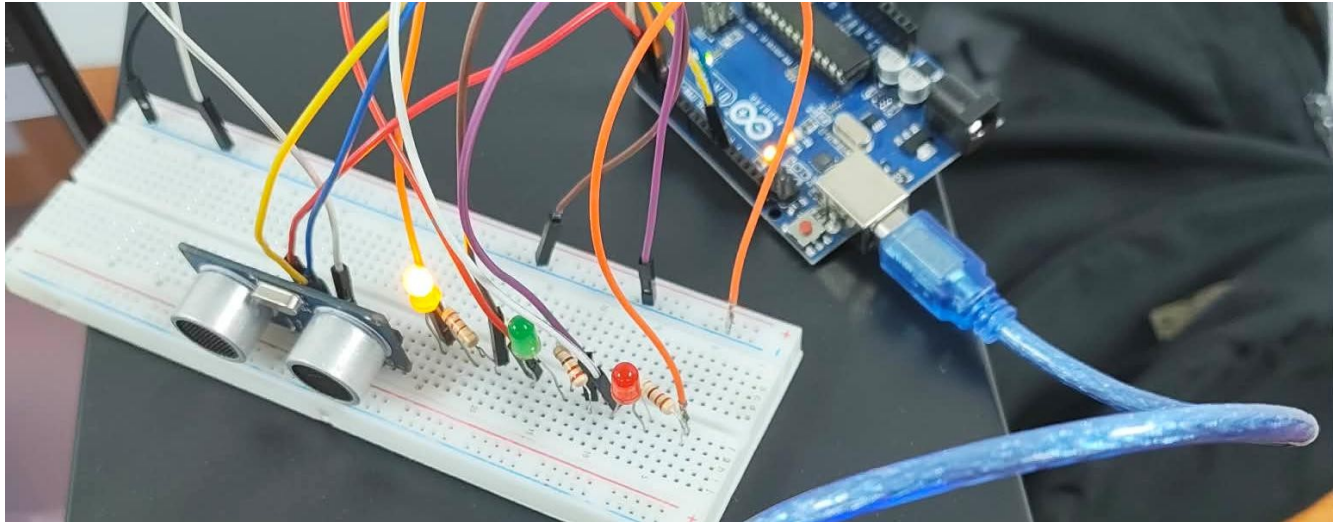


Figure 2.2: $30\text{ cm} < \text{Distance} \leq 50\text{ cm}$ **YELLOW LED ON only**

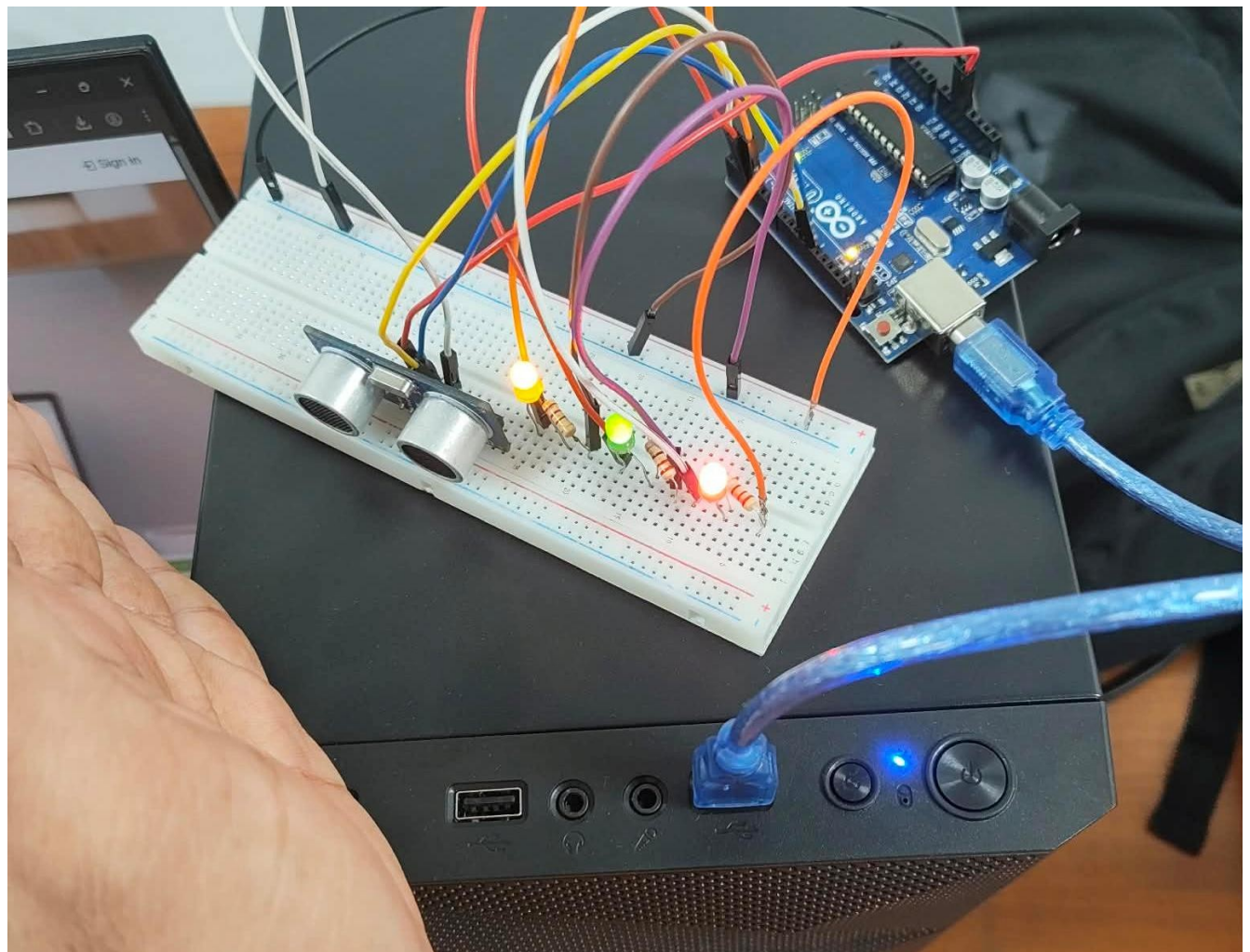


Figure 2.3: $\text{Distance} \leq 30\text{ cm}$ **ALL THREE LEDs ON (Red + Green + Yellow)**

Code/Program:

```
sketch_exp7Group02.ino
1 // define the pin numbers
2 const int trigPin = 11;
3 const int echoPin = 12;
4
5 // define LED pins (corrected logic: Green=Safe, Yellow=Caution, Red=Danger)
6 const int greenLedPin = 2; // Far/Safe
7 const int yellowLedPin = 3; // Medium distance
8 const int redLedPin = 4; // Close/Danger
9
10 // define variables
11 long duration;
12 float distance, distanceInches, distanceThreshold;
13
14 void setup() {
15     Serial.begin(9600); // Starts the serial communication
16     pinMode(trigPin, OUTPUT); // Sets the trigPin as an Output
17     pinMode(echoPin, INPUT); // Sets the echoPin as an Input
18     pinMode(greenLedPin, OUTPUT); // Sets the green LED pin as an Output
19     pinMode(yellowLedPin, OUTPUT); // Sets the yellow LED pin as an Output
20     pinMode(redLedPin, OUTPUT); // Sets the red LED pin as an Output
21 }
22
23 void loop() {
24     // Clears the trigPin
25     digitalWrite(trigPin, LOW);
```

Fig 3: Code for this program (1)

```
26     delayMicroseconds(2);
27
28     // Sets the trigPin on HIGH state for 10 microseconds
29     digitalWrite(trigPin, HIGH);
30     delayMicroseconds(10);
31     digitalWrite(trigPin, LOW);
32
33     // Reads the echoPin, returns the sound wave travel time in microseconds
34     duration = pulseIn(echoPin, HIGH);
35
36     // Calculating the distance
37     distance = (duration / 2.0) * 1e-6 * 340 * 100; // Distance in cm
38     distanceInches = distance / 2.54; // Distance in inches
39
40     // Prints the distance on the Serial Monitor
41     Serial.print("Distance = ");
42     Serial.print(distance);
43     Serial.print(" cm; ");
44     Serial.print("Distance = ");
45     Serial.print(distanceInches);
46     Serial.println(" inches");
47
48     // Set threshold distance to activate LEDs
49     distanceThreshold = 80;
50
```

Fig 4: Code for this program (2)


```

51 // Reset all LEDs to LOW
52 digitalWrite(greenLedPin, LOW);
53 digitalWrite(yellowLedPin, LOW);
54 digitalWrite(redLedPin, LOW);
55
56 if (distance > distanceThreshold) {
57     // Object is far - GREEN LED (Safe)
58     digitalWrite(greenLedPin, HIGH);
59 }
60 else if (distance <= distanceThreshold && distance > distanceThreshold - 30) {
61     // Object is at medium-far distance - GREEN LED
62     digitalWrite(greenLedPin, HIGH);
63 }
64 else if (distance <= distanceThreshold - 30 && distance > distanceThreshold - 50) {
65     // Object is at medium-close distance - YELLOW LED (Caution)
66     digitalWrite(yellowLedPin, HIGH);
67 }
68 else if (distance <= distanceThreshold - 50) {
69     // Object is very close - RED LED (Danger)
70     digitalWrite(redLedPin, HIGH);
71 }
72
73 delay(200); // Wait for 200 milliseconds
74 }

```

Fig 5: Code for this program (3)

Explanation:

This project uses an Arduino Uno to detect obstacles with an HC-SR04 ultrasonic sensor. The sensor works by sending out a sound pulse and timing how long it takes for the echo to return. The Arduino then uses the travel time to calculate how far away an object is.

To show the result, three different-colored LEDs provide a visual alert. A green light means the object is far away. A yellow light signals a medium distance. A red light warns that an object is very close.

In the code, specific Arduino pins are designated to control each LED and to listen for the sensor's signal. The simple decision-making logic, which activates the correct light based on the measured distance, illustrates a core concept in automated systems. This basic setup for sensing proximity and giving a clear warning has practical uses in areas like robotics, automated machinery, and safety devices.

Simulation Output Results:

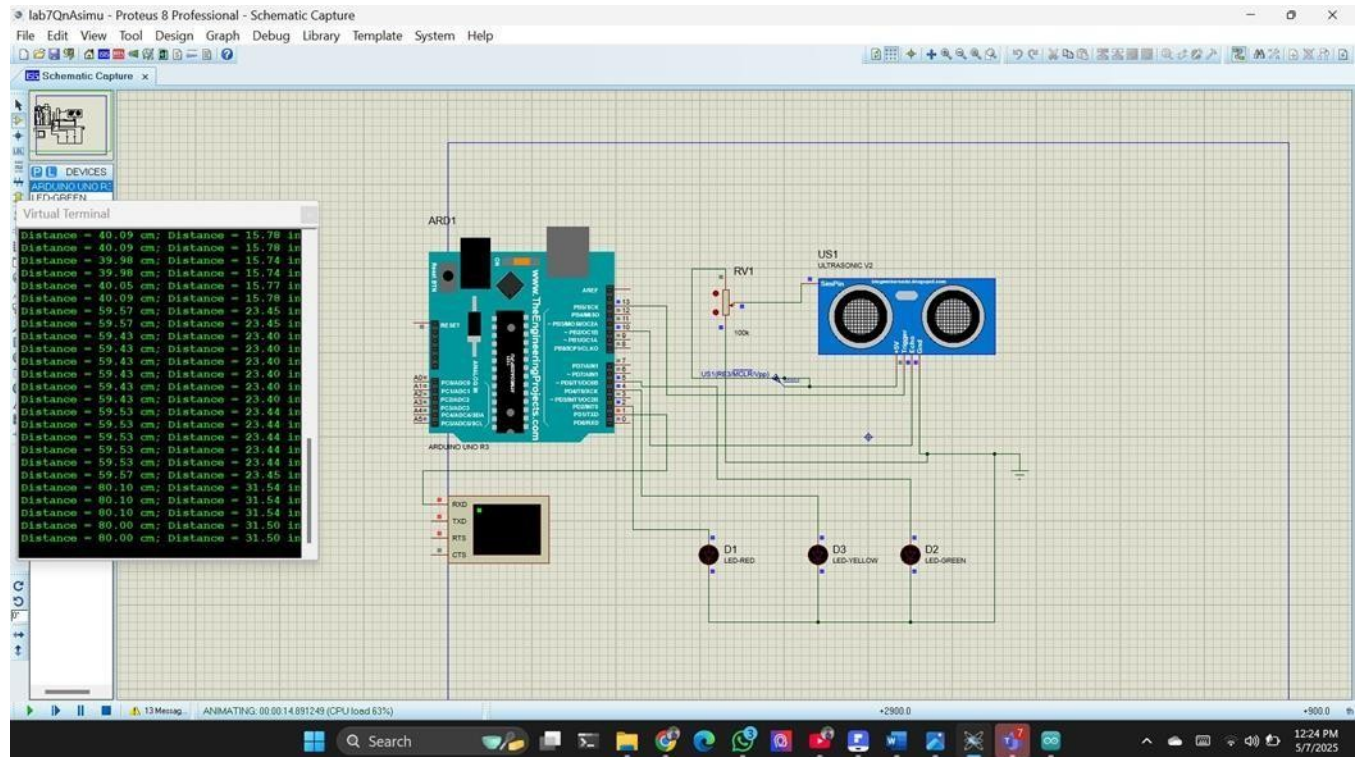


Fig 7: All LED OFF

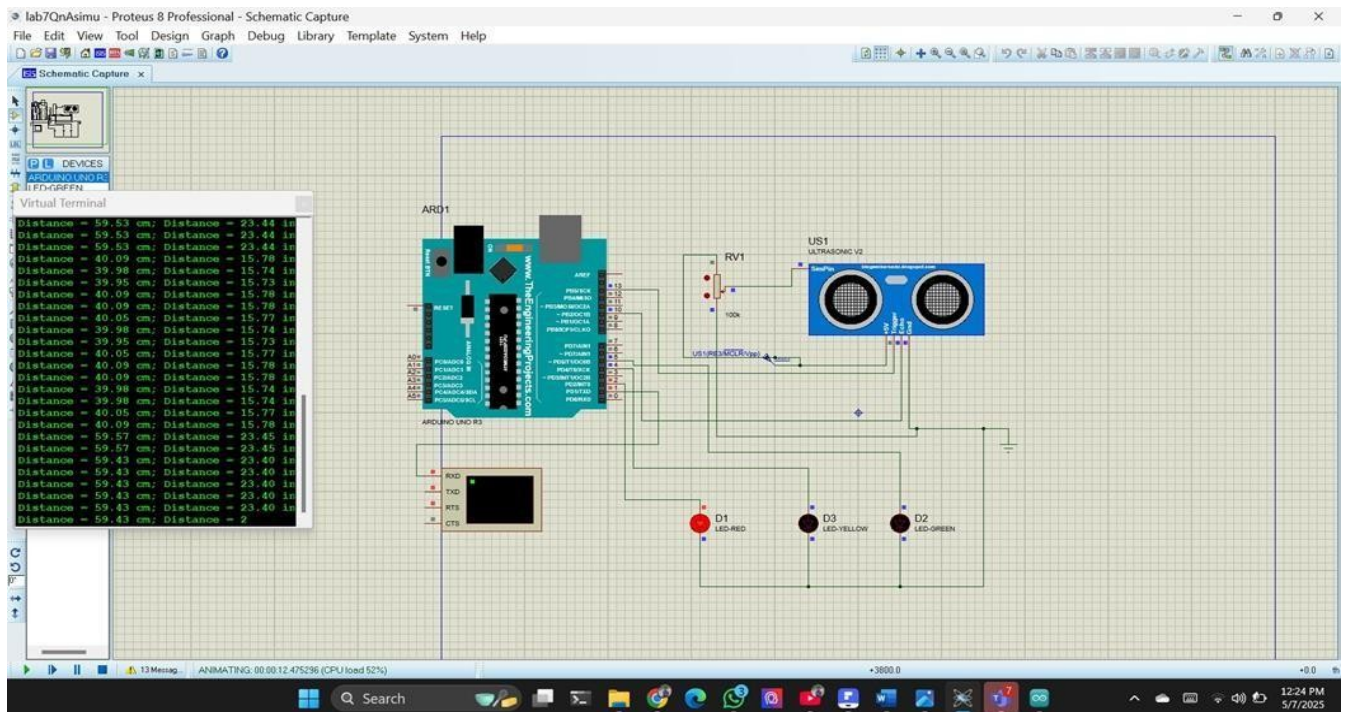


Fig 8: RED LED On

Discussion:

This hands-on project developed a functional proximity alert system using an Arduino Uno paired with an HC-SR04 sensor. The system calculates distance measurements, which are then translated into simple light signals: different LEDs illuminate to represent how near or far an object is.

All hardware wiring and programming logic were first rigorously tested within the Proteus simulation environment. This allowed for thorough debugging and verification of the design before any physical assembly.

By bridging theory and practice, the project demonstrates real-world applications in areas like robotics and automation. It also serves as an effective learning tool, building student competency in key skills such as sensor integration, conditional programming, and the essential practice of simulating complex circuits prior to physical prototyping.

References:

- [1] Arduino IDE, <https://www.arduino.cc/en/Main/Software> accessed on May 3, 2019.
- [2] Arduino and Proteus Library, <https://etechnophiles.com/add-simulate-ultrasonic-sensorproteus-2018-edition/> accessed on May 3, 2019.
- [3] Ultrasonic Distance Sensor in Arduino with the TinkerCad <https://www.instructables.com/id/UltrasonicDistanceSensor-Arduino-Tinkercad/> accessed on May 3, 2019.